A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-27960639})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K, the normalized generator σ_x, the pair (a', J) representing the torsion class e_j, the auxiliary divisor I', the compactly represented element t, a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1-\sigma_x)^2 I' + \operatorname{div}(t) + i_x(J) = 0, \qquad\qquad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$-basis of the ideal. For example, the ideal $J = \left(\begin{smallmatrix} 256 & 79 \\ 0 & 1 \end{smallmatrix}\right)$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}\,256 + \mathbf{Z}\,(79 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K. The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\mathrm{Cl}(K)$ (as HNF ideals). A character x on $\mathrm{Cl}(K)/5$ is recorded by its value vector $(x(\bar\kappa_1), x(\bar\kappa_2), x(\bar\kappa_3))$; thus $(1, 0, 0)$ is the character dual to $\bar\kappa_1$. The column number j records which basis element e_j of $\mathrm{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x. Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x.

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 6990160 = 0$, of discriminant -27960639. Class group invariants $(40 \quad 10 \quad 10)$, class number 4000; the three stored generator ideals (HNF) are $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 20 & 19 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 34 & 15 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a, column e_1

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s + 362)\, y^3 + (16s - 73459)\, y^2 + (83s + 544199)\, y + (1452s + 3527931)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 + 732*y^8 - 149071*y^7 + 8650029*y^6 - 272998314*y^5 +
6396477718*y^4 - 79132358538*y^3 + 150776944368*y^2 + 5525728284879*
y + 27188801985213
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad -1 \quad 1 \quad 1 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{59283}{68719476736} + (\frac{169}{1099511627776})\, s, \qquad J = \begin{pmatrix} 256 & 79 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $79542642483 = 3^2 \cdot 31 \cdot 285099077$:

$$\begin{pmatrix}
26514214161 & 24711311439 & 945739147 & 6564399060 & 15691855566 & 25477447456 & 22030121611 & 19221648915 & 16463720485 & 20387336740 \\
0 & 3 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 280 factors with signed exponents of absolute value up to 3334794; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 1 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 2: character a, column e_2

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s + 362)\, y^3 + (16s - 73459)\, y^2 + (83s + 544199)\, y + (1452s + 3527931)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 + 732*y^8 - 149071*y^7 + 8650029*y^6 - 272998314*y^5 +
6396477718*y^4 - 79132358538*y^3 + 150776944368*y^2 + 5525728284879*
y + 27188801985213
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad -1 \quad 1 \quad 1 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{31683}{128000000000} + (\frac{739}{10240000000000})\, s, \qquad J = \begin{pmatrix} 400 & 159 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 400$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $376422079103 = 7^2 \cdot 31 \cdot 247809137$:

$$\begin{pmatrix} 53774582729 & 19935366280 & 20929047601 & 4310316619 & 9272353290 & 21721542987 & 20124445464 & 51403496941 & 20730884886 & 31942101179 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 7 & 6 & 6 & 2 & 3 & 4 & 5 & 5 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 271 factors with signed exponents of absolute value up to 72509; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 1 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 3: character a, column e_3

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s + 362)\, y^3 + (16s - 73459)\, y^2 + (83s + 544199)\, y + (1452s + 3527931)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 + 732*y^8 - 149071*y^7 + 8650029*y^6 - 272998314*y^5 +
6396477718*y^4 - 79132358538*y^3 + 150776944368*y^2 + 5525728284879*
y + 27188801985213
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad -1 \quad 1 \quad 1 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{829281}{129023609628736} + (\frac{16435}{2064377754059776})\, s, \qquad J = \begin{pmatrix} 1156 & 423 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1156$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2304066410049 = 3 \cdot 53^2 \cdot 3109 \cdot 87943$:

$$\begin{pmatrix}
43472951133 & 16735763361 & 21740076201 & 15039705190 & 26875185104 & 10195685557 & 21693084067 & 5398109001 & 9225839602 & 33851083379 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 53 & 37 & 52 & 12 & 28 & 30 & 41 & 33 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 276 factors with signed exponents of absolute value up to 217212; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 2 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 4: character b, column e_1

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (s + 568)\, y^3 + (9s + 35476)\, y^2 + (-102s - 162448)\, y + (-5024s - 9961472)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 1138*y^8 + 69824*y^7 + 6917393*y^6 + 146561234*y^5 +
234242836*y^4 - 105931199816*y^3 - 1240056848096*y^2 + 1040246163916
8*y + 275716635574272
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0\quad 0\quad -1\quad -1\quad 0\quad 0\quad 0\quad -1\quad 0\quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{59283}{68719476736} + (\frac{169}{1099511627776})\, s, \qquad J = \begin{pmatrix} 256 & 79 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $15265292350896 = 2^4 \cdot 3^2 \cdot 7 \cdot 15144139237$:

$$\begin{pmatrix}
636053847954 & 31347976302 & 462408700251 & 480071415620 & 550539202422 & 500877787723 & 342721428841 & 404176897242 & 229310262107 & 106863193485 \\
0 & 2 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\
0 & 0 & 3 & 2 & 0 & 2 & 0 & 2 & 0 & 1 \\
0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 122 factors with signed exponents of absolute value up to 6370657; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0\quad 0\quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 5: character b, column e_2

Prescribed character vector $(0\quad 1\quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (s + 568)\, y^3 + (9s + 35476)\, y^2 + (-102s - 162448)\, y + (-5024s - 9961472)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 1138*y^8 + 69824*y^7 + 6917393*y^6 + 146561234*y^5 +
234242836*y^4 - 105931199816*y^3 - 1240056848096*y^2 + 1040246163916
8*y + 275716635574272
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0\quad 0\quad -1\quad -1\quad 0\quad 0\quad 0\quad -1\quad 0\quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{31683}{128000000000} + (\frac{739}{10240000000000})\, s, \qquad J = \begin{pmatrix} 400 & 159 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 400$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $83039392128 = 2^7 \cdot 3 \cdot 7^2 \cdot 11^2 \cdot 36473$:

$$\begin{pmatrix}
33701052 & 12956928 & 32850003 & 28745593 & 12048305 & 3569466 & 32663894 & 6251466 & 2479561 & 33422578 \\
0 & 4 & 3 & 1 & 3 & 0 & 2 & 1 & 1 & 2 \\
0 & 0 & 11 & 5 & 2 & 0 & 2 & 9 & 8 & 0 \\
0 & 0 & 0 & 28 & 8 & 4 & 7 & 24 & 23 & 16 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 143 factors with signed exponents of absolute value up to 12232723; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 6: character b, column e_3

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (s + 568)\, y^3 + (9s + 35476)\, y^2 + (-102s - 162448)\, y + (-5024s - 9961472)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 + 1138*y^8 + 69824*y^7 + 6917393*y^6 + 146561234*y^5 +
234242836*y^4 - 105931199816*y^3 - 1240056848096*y^2 + 1040246163916
8*y + 275716635574272
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{829281}{129023609628736} + (\frac{16435}{2064377754059776})\, s, \qquad J = \begin{pmatrix} 1156 & 423 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1156$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $4029051414528 = 2^{11} \cdot 3^5 \cdot 7^2 \cdot 17 \cdot 9719$:

$$\begin{pmatrix} 41636196 & 24894072 & 14318100 & 19348794 & 14751608 & 19620110 & 7134824 & 29705698 & 16603374 & 19720835 \\ 0 & 252 & 0 & 231 & 172 & 24 & 78 & 108 & 147 & 93 \\ 0 & 0 & 12 & 8 & 4 & 4 & 0 & 0 & 11 & 3 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 143 factors with signed exponents of absolute value up to 4510313; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 0 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 7: character c, column e_1

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 1588y^3 + (-12s + 6)\, y^2 + 268227y + (492s - 246)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 3176*y^8 + 3058198*y^6 + 154694052*y^4 - 10594082799*y^2 + 16
92066029724
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 1 \quad -2 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{59283}{68719476736} + (\frac{169}{1099511627776})\, s, \qquad J = \begin{pmatrix} 256 & 79 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $650590308480 = 2^7 \cdot 3 \cdot 5 \cdot 7 \cdot 337 \cdot 379^2$:

$$\begin{pmatrix} 107287320 & 86133056 & 30137932 & 71139459 & 97338409 & 77477726 & 104717966 & 3436235 & 67568773 & 34183847 \\ 0 & 758 & 308 & 695 & 613 & 502 & 534 & 421 & 68 & 346 \\ 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 108 factors with signed exponents of absolute value up to 54428864286325; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 4 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 8: character c, column e_2

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 1588y^3 + (-12s + 6)\, y^2 + 268227y + (492s - 246)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 3176*y^8 + 3058198*y^6 + 154694052*y^4 - 10594082799*y^2 + 16
92066029724
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 1 \quad -2 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{31683}{128000000000} + \left(\frac{739}{10240000000000}\right) s, \qquad J = \begin{pmatrix} 400 & 159 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 400$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9079560000000 = 2^9 \cdot 3^3 \cdot 5^7 \cdot 7 \cdot 1201$:

$$\begin{pmatrix} 25221000 & 7004340 & 22152150 & 5867540 & 18262256 & 18820385 & 17322853 & 17024845 & 24312696 & 18588219 \\ 0 & 30 & 0 & 10 & 10 & 15 & 4 & 28 & 13 & 15 \\ 0 & 0 & 30 & 0 & 28 & 15 & 16 & 24 & 5 & 29 \\ 0 & 0 & 0 & 20 & 8 & 0 & 2 & 16 & 17 & 18 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 10 & 0 & 4 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 108 factors with signed exponents of absolute value up to 36216091836435; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 2 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 9: character c, column e_3

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 1588y^3 + (-12s + 6)\, y^2 + 268227y + (492s - 246)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 3176*y^8 + 3058198*y^6 + 154694052*y^4 - 10594082799*y^2 + 16
92066029724
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 1 \quad -2 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{829281}{129023609628736} + \left(\frac{16435}{2064377754059776}\right) s, \qquad J = \begin{pmatrix} 1156 & 423 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1156$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1735776000000 = 2^{11} \cdot 3^3 \cdot 5^6 \cdot 7^2 \cdot 41$:

$$\begin{pmatrix} 688800 & 630000 & 92600 & 302610 & 362104 & 465493 & 333209 & 322510 & 143316 & 273144 \\ 0 & 2100 & 1920 & 1020 & 917 & 2058 & 1051 & 1700 & 58 & 511 \\ 0 & 0 & 20 & 0 & 17 & 14 & 12 & 9 & 9 & 6 \\ 0 & 0 & 0 & 30 & 1 & 6 & 17 & 15 & 8 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 108 factors with signed exponents of absolute value up to 32870510982944; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 2598y^3 + (42s - 21)\, y^2 - 2016090y + (-5742s + 2871)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 5196*y^8 + 2717424*y^6 + 1855038159*y^4 + 693069116202*y^2 +
230469509407599
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 2 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{59283}{68719476736} + (\frac{169}{1099511627776})\, s, \qquad J = \begin{pmatrix} 256 & 79 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1214806835889 = 3^2 \cdot 59 \cdot 1459 \cdot 1568041$:

$$\begin{pmatrix} 404935611963 & 256845522633 & 294457973886 & 152352640584 & 56007016542 & 384904553439 & 402136852953 & 53055595063 & 243187107794 & 91393781549 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \quad 1 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 2598y^3 + (42s - 21)\, y^2 - 2016090y + (-5742s + 2871)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 5196*y^8 + 2717424*y^6 + 1855038159*y^4 + 693069116202*y^2 +
230469509407599
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 2 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{31683}{128000000000} + \left(\frac{739}{10240000000000}\right) s, \qquad J = \begin{pmatrix} 400 & 159 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 400$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1441803466593 = 3^2 \cdot 7 \cdot 29 \cdot 789164459$:

$$\begin{pmatrix} 480601155531 & 76775129904 & 70478139063 & 236098937568 & 241724469483 & 94953026049 & 321152736534 & 263214761 & 393255030061 & 258284131067 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 4 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 2598y^3 + (42s - 21)\,y^2 - 2016090y + (-5742s + 2871)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 5196*y^8 + 2717424*y^6 + 1855038159*y^4 + 693069116202*y^2 +
230469509407599
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 2 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{829281}{129023609628736} + \left(\frac{16435}{2064377754059776}\right) s, \qquad J = \begin{pmatrix} 1156 & 423 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1156$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $56460574413 = 3^2 \cdot 113 \cdot 271 \cdot 204859$:

$$\begin{pmatrix}
18820191471 & 3108702567 & 1778812275 & 392202774 & 750720840 & 18085079064 & 17040754600 & 4754358896 & 7385149305 & 33153260 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 & 2 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 2 factors with signed exponents of absolute value up to 1; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 4 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 1115y^3 + (22s - 11)\, y^2 - 1513851y + (-6114s + 3057)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 2230*y^8 - 1784477*y^6 + 7349589*y^4 + 411280034895*y^2 + 261
299133654111
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{59283}{68719476736} + \left(\frac{169}{1099511627776}\right) s, \qquad J = \begin{pmatrix} 256 & 79 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $7609111923 = 3 \cdot 7 \cdot 17^3 \cdot 73751$:

$$\begin{pmatrix} 447594819 & 323377740 & 31761240 & 362185791 & 157444548 & 228474612 & 219400899 & 279871745 & 156571347 & 133648339 \\ 0 & 17 & 4 & 4 & 12 & 12 & 9 & 15 & 15 & 4 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 184 factors with signed exponents of absolute value up to 31329896; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \quad 4 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 1115y^3 + (22s - 11)\, y^2 - 1513851y + (-6114s + 3057)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 2230*y^8 - 1784477*y^6 + 7349589*y^4 + 411280034895*y^2 + 261
299133654111
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{31683}{128000000000} + \left(\frac{739}{10240000000000}\right) s, \qquad J = \begin{pmatrix} 400 & 159 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 400$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3913128132039 = 3^3 \cdot 7 \cdot 17^2 \cdot 859 \cdot 83401$:

$$\begin{pmatrix}
434792014671 & 218554853388 & 252064442937 & 338231574325 & 145429686276 & 360524388978 & 62681663554 & 41862296526 & 16696504278 & 403417226380 \\
0 & 3 & 0 & 1 & 2 & 0 & 1 & 1 & 0 & 1 \\
0 & 0 & 3 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 171 factors with signed exponents of absolute value up to 6272277; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \quad 4 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 1115y^3 + (22s - 11)\,y^2 - 1513851y + (-6114s + 3057)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 2230*y^8 - 1784477*y^6 + 7349589*y^4 + 411280034895*y^2 + 261
299133654111
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{829281}{129023609628736} + \left(\frac{16435}{2064377754059776}\right) s, \qquad J = \begin{pmatrix} 1156 & 423 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1156$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $32998704777 = 3^7 \cdot 17 \cdot 43 \cdot 20641$:

$$\begin{pmatrix}
45265713 & 21095646 & 5530470 & 19601982 & 30756711 & 43359900 & 10629264 & 18497777 & 17128620 & 16999861 \\
0 & 3 & 0 & 0 & 0 & 0 & 0 & 1 & 2 & 0 \\
0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 1 & 2 \\
0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 & 1 & 1 \\
0 & 0 & 0 & 0 & 3 & 0 & 0 & 2 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 & 1 & 2 \\
0 & 0 & 0 & 0 & 0 & 0 & 3 & 1 & 2 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 171 factors with signed exponents of absolute value up to 46815; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s - 2295)\, y^3 + (43s + 61869)\, y^2 + (-650s - 1223686)\, y + (6176s + 14314576)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 4590*y^8 + 128372*y^7 + 9687677*y^6 - 854209674*y^5 +
31433246216*y^4 - 694334084272*y^3 + 9936525679056*y^2 - 9117255106
1408*y + 471620997997312
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad 0 \quad -1 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{59283}{68719476736} + (\frac{169}{1099511627776})\, s, \qquad J = \begin{pmatrix} 256 & 79 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 256$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $7030651379904 = 2^6 \cdot 3^6 \cdot 47 \cdot 3206197$:

$$\begin{pmatrix}
5424885324 & 2341675500 & 118812888 & 604757892 & 4986909483 & 3806015442 & 1071927618 & 713937756 & 2312544613 & 2131055808 \\
0 & 3 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\
0 & 0 & 6 & 0 & 0 & 0 & 2 & 3 & 2 & 0 \\
0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 & 2 \\
0 & 0 & 0 & 0 & 0 & 6 & 2 & 4 & 5 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 152 factors with signed exponents of absolute value up to 118950801021710; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s - 2295)\, y^3 + (43s + 61869)\, y^2 + (-650s - 1223686)\, y + (6176s + 14314576)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 4590*y^8 + 128372*y^7 + 9687677*y^6 - 854209674*y^5 +
31433246216*y^4 - 694334084272*y^3 + 9936525679056*y^2 - 9117255106
1408*y + 471620997997312
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad 0 \quad -1 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{31683}{128000000000} + \left(\frac{739}{10240000000000}\right) s, \qquad J = \begin{pmatrix} 400 & 159 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 400$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3645932083632 = 2^4 \cdot 3^3 \cdot 43 \cdot 47 \cdot 4175981$:

$$\begin{pmatrix}
50637945606 & 34660801230 & 20241723450 & 35456776712 & 14228942801 & 10850098676 & 39764970674 & 26253320114 & 25646411235 & 31788065096 \\
0 & 6 & 0 & 0 & 4 & 0 & 3 & 0 & 0 & 5 \\
0 & 0 & 6 & 2 & 3 & 0 & 0 & 0 & 0 & 2 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 156 factors with signed exponents of absolute value up to 116731856301755; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 2 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s - 2295)\, y^3 + (43s + 61869)\, y^2 + (-650s - 1223686)\, y + (6176s + 14314576)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 4590*y^8 + 128372*y^7 + 9687677*y^6 - 854209674*y^5 +
31433246216*y^4 - 694334084272*y^3 + 9936525679056*y^2 - 9117255106
1408*y + 471620997997312
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad 0 \quad -1 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{829281}{129023609628736} + (\frac{16435}{2064377754059776})\, s, \qquad J = \begin{pmatrix} 1156 & 423 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1156$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $53022790278 = 2 \cdot 3^2 \cdot 43 \cdot 47 \cdot 1457551$:

$$\begin{pmatrix}
17674263426 & 16538513577 & 10961985960 & 10802125878 & 11760885601 & 8973251227 & 15491889622 & 9928285122 & 16879405308 & 3651000862 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 3 & 1 & 0 & 0 & 1 & 2 & 1 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 159 factors with signed exponents of absolute value up to 54316773528934; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 5$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.